Volkan Kilinc, Ph.D.

Materials Scientist & AI Product Builder

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PROFILE

Materials scientist (Ph.D., Aix-Marseille) with 8 years of R&D in France, Japan and Switzerland, specialised in DNA nanomaterials and transistor-based biosensors. 10 peer-reviewed papers (5 first-author), 2 patent applications, and competitive fellowships from JSPS, TÜBİTAK and the European Commission (MSCA Seal of Excellence). Now an AI Content Domain Expert (Physics) at Mercor and founder of ReviewMetric, an AI peer-review platform built end to end and launched in November 2026.

KEY ACHIEVEMENTS

TÜBİTAK 2232-A Fellowship for Outstanding Researchers · 2026
24–36-month fellowship with a 4.7M TL initial research grant to run SEMANTIC-FET (DNA-FET arrays, AI drift correction, digital twins) at BTS Labs, Türkiye.

MSCA Postdoctoral Fellowship — Seal of Excellence · 2026
For AXON-DNA, an AI-indexed DNA-FET array for DNA data retrieval (proposed host: Empa, Switzerland).

JSPS Postdoctoral Fellowship · 2022
24-month stipend plus a ¥2.3M KAKENHI research grant to pioneer DNA field-effect transistors for DNA data storage.

Founder of ReviewMetric · 2026 launch
Six AI referees and a handling editor review manuscripts before submission. Inventor on 2 Japanese patent applications.

EXPERIENCE

AI Content Domain Expert (Physics) — Mercor — 09/2026 – present
Remote · joined as AI Content Verifier (05/2026 – 08/2026)

- Improve frontier AI models on physics and frontier-science reasoning, and on professional-grade presentation generation.
- Write and verify expert tasks, rubrics and evaluations in physics, materials science and chemistry.

Founder & Developer — ReviewMetric (formerly Open AI Science) — 06/2025 – present
Paris · reviewmetric.it.com

- Designed and shipped, alone, a multi-agent AI platform that reviews manuscripts for novelty, methods, citations, figures and writing, with each finding pinned to its paragraph.
- Built the full stack: Python/Flask API, async NLP workers, PostgreSQL (Supabase), Render cloud, WebSocket live dashboards, Stripe pay-as-you-go billing.
- Published an open-source MCP server connecting LLMs to the platform; built the ReviewMetric Atlas, a map of research frontiers.

JSPS Postdoctoral Fellow, then Invited Researcher — NIMS — 04/2022 – 04/2025
Tsukuba, Japan · host: Dr J. P. Hill

- Invented a reusable nanoporous DNA-FET that picks one DNA "search tag" out of 9 near-identical sequences at sub-femtomolar levels in ~9 min → first-author paper (*Adv. Sensor Res.* 2024) + JP patent application.
- Created "DNA-pods", sequence-independent DNA microrods that unfold above 60 °C to release their cargo; showed doxorubicin delivery to cancer-cell nuclei → first-author paper (*Chem. Mater.* 2026).
- Ran the projects end to end: idea, funding, experiments, writing, IP.

CTO & Co-founder — Gensor — 06/2021 – 06/2023
Paris

- Co-founded a deep-tech start-up commercialising a gene-FET virus test; selected for the MassChallenge global accelerator.

Technology Transfer Project Manager — CNRS Innovation — 01/2021 – 01/2022
Paris

- Assessed the market potential of early-stage (TRL 1–2) lab inventions; built valorisation and IP roadmaps.
- Coached researchers on EIC Transition and ERC Proof-of-Concept applications.

Ph.D. Researcher — CINaM (CNRS / Aix-Marseille Univ.) 03/2017 – 05/2020
Marseille, France

- Built a transistor sensor detecting cesium in seawater for radioactive-pollution monitoring (France–Japan ANR project SenCei): synthesised selective probes, designed lipid/polymer interfaces → *ACS AMI* 2019, *Sens. Actuators B* 2022 (co-first), JP patent application.

Research Intern — ABB Corporate Research (CH) · Nagaoka Univ. of Technology (JP) 2015 – 2016

- Ageing and failure analysis of insulating materials (GC-MS, FTIR, SEM, XRD) · cross-coupling synthesis.

EDUCATION

Ph.D. 2020	Nanoscience & Condensed Matter — Aix-Marseille Université
M.Sc. 2016	Polymer Science, magna cum laude — Université Paris-Est Créteil
B.Sc. 2014	Biology & Chemistry, magna cum laude — Université Paris-Est Créteil · Erasmus year, Universidad de Murcia

SKILLS

Science	DNA & lipid soft materials, surface functionalisation, FET biosensors, drug delivery, organic synthesis; SEM, XRD, FTIR, GC-MS, chromatography, spectroscopy
AI & software	Multi-agent LLM systems, MCP, Python/Flask, React, React Native, Node.js, PostgreSQL, Supabase, Render, WebSockets, Stripe
Leadership	Grant writing (JSPS, MSCA, TÜBİTAK, EIC, ERC), patent drafting, start-up creation, project management
Languages	French (native), Turkish (native), English (fluent, working language), Spanish (conversational)

SELECTED PUBLICATIONS — 10 total · 5 first-author · 3 corresponding (*) · full list on Google Scholar

1. **Kilinc V.***, Sutrisno L., Henzie J., Picheau E., Yamauchi Y., Ariga K., Hill J.P.* (2026). Hierarchical self-assembly of disulfide-linked single-stranded DNA into stimuli-responsive pods. *Chemistry of Materials* 38(4), 1968–1979. doi
2. **Kilinc V.***, Hayakawa R., Yamauchi Y., Wakayama Y., Hill J.P.* (2024). Nanoporous DNA field-effect transistor with potential for random-access memory applications: a selectivity performance evaluation. *Advanced Sensor Research* 3(5), 2300176. doi
3. **Nguy T.P., Kilinc V.** (co-first), Hayakawa R., Henry-de-Villeneuve C., Raimundo J.-M.*, Wakayama Y.*, Charrier A.* (2022). Affinity-driven ion-exchange EG-OFET sensor for high selectivity and low limit of detection of cesium in seawater. *Sensors and Actuators B* 351, 130956. doi
4. **Kilinc V.**, Henry-de-Villeneuve C., Nguy T.P., Wakayama Y., Charrier A.*, Raimundo J.-M.* (2019). Novel and innovative interface as potential active layer in Chem-FET sensor devices for the specific sensing of Cs^+. *ACS Applied Materials & Interfaces* 11(50), 47635–47641. doi

HONOURS

15th HOPE Meeting with Nobel Laureates, Kyoto (2024) · Poster Award, Giornate Italo-Francesi di Chimica (2018) · 10+ international conferences, 5 oral talks (1 invited).

References: Dr Jonathan P. Hill (NIMS, Japan) · Dr Anne Charrier (CNRS, CINaM, France) · Prof. Jean-Manuel Raimundo (CINaM, France).